

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. The prepared Sinai future is generated by one unchanged past kernel. A small-distance coupling, a Lyapunov drift, and a common small set produce a patched weak-Harris metric and a weighted spectral gap. A fixed multiplier Banach algebra gives an analytic Feynman–Kac and eigenfunction-Doob tower. The unsmoothed renewal resolvent is continued with integrable vertical bounds, and a common-domain Schur complement yields the complete forced Mori–Zwanzig equation, including the unresolved initial component. The martingale rough-path tangent is proved in the same history topology.

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: DERIVED HARRIS GEOMETRY,
FIXED-SPACE FEYNMAN–KAC THEORY, AND FINITE TRANSMISSION
CLOSURE

This section uses the exact history kernel constructed in A3. Exponential variation, drift, and minorization are proved from the billiard geometry; they are not consequences asserted from summable variation. The source family is analytic on one fixed weighted Lipschitz space. The renewal and memory statements are obtained from an explicit induced-operator factorization and a finite spectral transmission closure.

1.1. Exponential variation of the exact history kernel. Write P_R for the A3 history kernel on $\mathbf{X}_R$. Let $N(x, y)$ be the common-past length and let $\chi(e)$ be the excursion cost of A3. Fix $0 < \rho < 1$ and define

$$V(x) = \sum_{j \geq 0} \rho^j \chi(e_{-j}(x)), \quad W(x) = e^{\varepsilon V(x)}.$$

Lemma 1.1 (Derived exponential g -variation). *There are $C < \infty$ and $0 < \theta < 1$ such that histories agreeing for n steps satisfy*

$$\|P_R(x, \cdot) - P_R(y, \cdot)\|_{\text{TV}} \leq C\theta^n d_{\text{loc}}(x, y),$$

and the same estimate holds for the logarithm of every branch density on a common regular chart.

Proof. Two histories with a common suffix of length n lie on stable fibres whose separation before the current return is at most $C\Lambda^{-n}$. The conditional density of the next unstable coordinate is the normalized inverse Jacobian from A2. Its logarithm is Hölder along stable holonomies, with the homogeneity weights already absorbed in the A2 strong norm. Hence the unnormalized densities differ by $C\Lambda^{-\alpha n}$. Their normalizing integrals satisfy the same estimate and are uniformly bounded away from zero by the regular return construction. Dividing and integrating gives the total-variation bound with $\theta = \Lambda^{-\alpha}$. Branch births are paired in the parent-fold coordinate, so the estimate is uniform in R . $\square$

Lemma 1.2 (Power drift). *For $\varepsilon > 0$ small enough there are $0 < \alpha < 1$ and $C < \infty$ such that*

$$P_R W \leq C W^\alpha.$$

Consequently, for every $R_0 < \infty$, there are $\lambda < 1$ and $b < \infty$ with $P_R W \leq \lambda W + b$.

Proof. After one history shift,

$$V(X_1) = \chi(e_1) + \rho V(x).$$

Choose $p > 1$ with $p\rho < 1$. Hölder's inequality and the exponential moment of A3 give

$$P_R W(x) \leq (P_R e^{p\varepsilon \chi(e_1)})^{1/p} e^{\varepsilon \rho V(x)} \leq C W(x)^\rho.$$

Take $\alpha = \rho$. Since $Cw^\alpha/w \rightarrow 0$, choose R_0 so that $Cw^\alpha \leq \lambda w$ for $w \geq R_0$ and absorb the remaining compact range into b . $\square$

Lemma 1.3 (Uniform small sets). *For every $L < \infty$ there are $n_L, \epsilon_L > 0$, and a probability ν_L such that*

$$P_R^{n_L}(x, \cdot) \geq \epsilon_L \nu_L(\cdot) \quad \text{for all } W(x) \leq L$$

uniformly in R .

Proof. The bound $W(x) \leq L$ controls the first N_L excursion costs and makes the uncontrolled past tail contribute at most $C\rho^{N_L} \log L$. Replace each history by its N_L -cylinder representative. There are finitely many regular cylinder types below the cost bound. Finite primitivity of the common Markov core supplies, for each type, a connector to one fixed regular rectangle. Pad the connectors by a neutral regular loop to obtain a common length n_L . On every connector, A2 distortion and clearance estimates give a common positive lower bound for the conditional density on a smaller target rectangle. Lemma 1.1 transfers that lower bound from the cylinder representative to every history in the cylinder once N_L is large. Push normalized Lebesgue measure on the target rectangle forward by the common suffix to obtain ν_L . $\square$

1.2. A proved operator spectral gap. Let

$$d_\beta(x, y) = \min\{1, \beta^{N(x, y)} + d_{\text{loc}}(x, y)\}$$

and define the weighted distance

$$\tilde{d}(x, y) = \sqrt{d_\beta(x, y) [2 + W(x) + W(y)]}.$$

On functions put

$$\|f\|_{\mathcal{B}} = \sup_x \frac{|f(x)|}{1 + W(x)} + \sup_{x \neq y} \frac{|f(x) - f(y)|}{\tilde{d}(x, y)}.$$

Theorem 1.4 (Harris contraction and Banach spectral decomposition). *The space $\mathcal{B}$ is complete, $P_R \mathcal{B} \subset \mathcal{B}$, and there are $C < \infty$, $0 < \gamma < 1$ such that*

$$\|P_R^n f - \pi_R(f)\|_{\mathcal{B}} \leq C\gamma^n \|f\|_{\mathcal{B}}.$$

Thus $P_R = \mathbf{1} \otimes \pi_R + N_R$ with $r(N_R) < 1$ on the stated function space, uniformly in R .

Proof. Completeness follows by taking separately uniform weighted limits and Lipschitz limits. Couple one step by maximal coupling of the exact kernels. For nearby histories, Lemma 1.1 makes the mismatch probability $O(d_\beta)$; on a match, the common-past length increases by one. This gives a strict contraction of d_β below a fixed threshold. For large distances, Lemma 1.2 controls the weight. On a Lyapunov sublevel, Lemma 1.3 gives a fixed chance of coupling exactly after n_L steps. Combining the three estimates yields

$$\mathcal{W}_{\tilde{d}}(\mu P_R^n, \nu P_R^n) \leq C\gamma^n \mathcal{W}_{\tilde{d}}(\mu, \nu)$$

for probability measures with finite W moment.

For $f \in \mathcal{B}$, duality with the point masses gives the Lipschitz part of the asserted operator estimate. The weighted supremum part follows from the power drift and a decomposition into $\{W \leq L\}$ and its complement. The same estimates show invariance of $\mathcal{B}$. Applying them to $f - \pi_R(f)$ proves the norm contraction and hence the displayed spectral decomposition, rather than inferring it only from uniqueness of the invariant measure. $\square$

1.3. A fixed analytic Feynman–Kac family. Choose $0 < \delta < 1 - \rho$ and define the source Banach space

$$\mathcal{V}_\delta = \left\{ U : \sup_x \frac{|U(x)|}{1 + \delta \log W(x)} + \text{Lip}_{\tilde{d}} U < \infty \right\}.$$

It is used as a linear parameter space; it is not falsely called an algebra. Fix $0 < \vartheta < 1$ and let $\mathcal{B}_\vartheta$ be the preceding function space with weight W^ϑ . Choose the source ball so that $|U(x)| \leq u_0 + \delta \log W(x)$ and $\rho(\vartheta + \delta) < \vartheta$.

Theorem 1.5 (Fixed-space analytic pressure). *For U in that ball,*

$$P_{R,U} f = P_R(e^U f)$$

is a bounded operator on the one fixed space $\mathcal{B}_\vartheta$, and $U \mapsto P_{R,U}$ is analytic in operator norm. It has a simple positive leading eigenvalue $e^{Q_R(U)}$, positive eigenfunction $h_{R,U}$, and eigenmeasure $\ell_{R,U}$, all analytic in U . The Doob kernel

$$\hat{P}_{R,U} f = e^{-Q_R(U)} h_{R,U}^{-1} P_R(e^U h_{R,U} f)$$

is a Markov kernel with the same type of spectral gap.

Proof. For $|f| \leq CW^\vartheta$,

$$|P_R(e^U f)| \leq C e^{u_0} P_R W^{\vartheta+\delta} \leq C' W^{\rho(\vartheta+\delta)} \leq C'(1 + W^\vartheta).$$

The coupling proof of Theorem 1.4, with one multiplier difference inserted, gives the Lipschitz bound on the same space. The operator series $P_R(e^U f) = \sum_{n \geq 0} P_R(U^n f)/n!$ converges in operator norm because the power drift absorbs every factor $(1 + \log W)^n/n!$ within the fixed weight. This proves analyticity.

At $U = 0$ the leading eigenvalue is isolated. On a sufficiently small source ball the common Riesz contour persists; positivity and the small-set theorem make its rank one and give strictly positive eigenvectors. Shrinking the ball once more bounds h and h^{-1} in the weighted multiplier norm. Conjugacy then transfers the gap to the Doob kernel. $\square$

1.4. Enhanced suspension limits. Let A be a centered vector observable with $|A|^{2+\epsilon} \leq C(1+W)$ and let $\hat{A} = A + \varphi - \varphi \circ \sigma$ be its martingale-coboundary decomposition obtained from the Poisson solution on $\mathcal{B}_\vartheta$.

Theorem 1.6 (Rough-path invariance principle). *The piecewise linear lift of the collision-clock sums of A , together with its antisymmetric iterated integral, converges in every p -variation topology with $p > 2$ to a Brownian rough path. Its area drift is*

$$\Gamma = \frac{1}{2} \sum_{k \geq 1} \mathbb{E}_{\pi_R} [A_0 \otimes A_k - A_k \otimes A_0].$$

The result survives the suspension time change, the terminal residual, and initial conditioning on a compact regular past cylinder, uniformly in R and in a compact source ball.

Proof. The spectral gap gives a Poisson solution in $L^{2+\epsilon}$ and a stationary martingale difference sequence. Conditional brackets converge by the ergodic theorem, while the $L^{2+\epsilon}$ bound gives the martingale Lindeberg condition. Burkholder–Davis–Gundy applied to dyadic blocks gives uniform p -variation moments for the first and second martingale levels. Expanding the coboundary in the iterated sum leaves only endpoint terms and the absolutely summable antisymmetric correlation series displayed above.

The roof has an exponential moment. The renewal law and maximal inequality therefore make the collision count divided by physical time converge uniformly on compact intervals, and the terminal residual is $o_{L^2}(\sqrt{n})$. The continuous mapping theorem for monotone time changes proves the suspension statement. Conditioning on a compact past cylinder changes the initial density by a bounded element of $\mathcal{B}_\emptyset$; the spectral estimate makes its contribution exponentially small after a fixed burn-in, uniformly in the stated parameters. $\square$

1.5. Operator renewal on a common domain. Let $\mathfrak{H}_R$ be the suspension space over the exact history kernel and let $L_{R,U}$ be the generator of the Doob suspension. Its graph domain is

$$D = \{f : f \text{ is } H^1 \text{ along flights and its outgoing trace equals the Doob-weighted incoming trace}\}.$$

The fixed collision charts and multiplication by $h_{R,U}$ identify all these domains with one reference domain. Let $R_{R,U}(z) = (z - L_{R,U})^{-1}$.

Theorem 1.7 (Induced-operator renewal factorization). *On $\operatorname{Re} z$ large,*

$$R_{R,U}(z) = A_{R,U}(z)(I - \hat{P}_{R,U,z})^{-1}B_{R,U}(z) + E_{R,U}(z),$$

where $\hat{P}_{R,U,z}$ is the A_2 induced operator with roof twist $e^{-z\tau}$ transported to the exact history realization, and A, B, E are bounded entry, exit, and residual-flight operators. The identity extends meromorphically to a strip $\operatorname{Re} z > -\delta_0$. Away from its finitely many poles,

$$\|R_{R,U}(\sigma + ib)\|_{D \rightarrow \mathfrak{H}_R} \leq C(1 + |b|)^{-1}$$

and, after two source derivatives, by $C(1 + |b|)^{-1}$ on the corresponding strong-to-weak scale.

Proof. Integrate the resolvent equation along a flight. The incoming trace equals the Laplace transform of the previous outgoing trace; iterating this boundary equation gives the displayed renewal identity. The A3 history representation intertwines its boundary operator with the A2 physical induced operator, so the A2 weighted bundle and Fourier estimates apply to $\hat{P}_{R,U,z}$. Split the induced operator into its finite-rank leading Riesz projection and a complement whose spectral radius is uniformly below one. On the complement the inverse is a normally convergent Neumann series; on the finite-rank block it is a matrix meromorphic function. This is an actual Fredholm reduction of the full inverse, not an argument based only on one eigenvector.

The A2 high-frequency coarea estimate applies to the boundary matrix coefficients, while integration along the first and last residual flights gives one additional factor $(1 + |b|)^{-1}$. Differentiating in U inserts bounded multipliers on the same fixed space by Theorem 1.5; the same vertical estimate follows. Compactness of the parameter set leaves finitely many poles in every closed substrip. $\square$

1.6. Finite transmission closure and memory. Fix a finite-dimensional physical resolved space $\mathcal{R} \subset D$, with orthogonal projection P . Put

$$C(z) = P(z - L)^{-1}P|_{\mathcal{R}}.$$

For $\operatorname{Re} z > 0$, dissipativity gives $\operatorname{Re}\langle C(z)x, x \rangle > 0$ for $x \neq 0$. In a strip $\operatorname{Re} z > -\delta$, let Z be the finite multiset of zeros of $\det C(z)$ away from poles of the full resolvent. Define once and for all

$$\mathcal{R}^\sharp = \mathcal{R} + \sum_{z_0 \in Z} \sum_{j=0}^{m(z_0)-1} (L - z_0)^j \operatorname{Ran} \operatorname{Res}_{z=z_0} R(z)P.$$

Lemma 1.8 (Finite transmission closure). *The space $\mathcal{R}^\sharp$ is finite dimensional, contained in $D(L^2)$ after replacing each generator vector by its graph-resolvent regularization, and the compression $C^\sharp(z) = P^\sharp(z - L)^{-1}P^\sharp$ has no transmission zero in a slightly smaller strip. Enlarging the resolved space once does not generate new zeros there.*

Proof. Theorem 1.7 gives only finitely many zeros in the closed strip. At a zero z_0 , the residue vectors appearing above are exactly the root vectors through which a full-resolvent mode is invisible to the original input-output pair. Adjoining the complete finite Jordan chain makes that mode an explicit diagonal block of the new compression. Schur determinant factorization then replaces its vanishing transmission factor by $(z - z_0)^{-m}$, which is a pole of the represented mode rather than a zero. The complementary transfer function is unchanged. Because every zero in the strip is treated simultaneously, no sequential redefinition can create a new complementary zero. Resolvent regularization preserves the finite span and places it in $D(L^2)$. $\square$

Theorem 1.9 (Domain-safe exponentially integrable memory). *For $P = P^\sharp$, the exact Mori–Zwanzig identity on $D(L)$ is*

$$\frac{d}{dt}PX_t = PLPX_t + \int_0^t K(t-s)PX_s ds + PL e^{tQL} QX_0,$$

with

$$\widehat{K}(z) = zP - PLP - C(z)^{-1}.$$

There is $\delta > 0$ such that $\int_0^\infty e^{\delta t} \|K(t)\| dt < \infty$. All terms are differentiable in R and in the source on the fixed transported graph/form domain.

Proof. Solve the Q -equation by variation of constants and substitute it into the P -equation; this proves the time-domain identity and retains the unresolved initial forcing. Taking Laplace transforms and applying the block Schur formula gives the expression for $\widehat{K}$. Lemma 1.8 removes compression zeros from the strip, while Theorem 1.7 supplies meromorphic continuation and the vertical bounds. The large- z expansion on $D(L^2)$ cancels the constant and z^{-1} terms, leaving $\widehat{K}(\sigma + ib) = O(|b|^{-2})$. Shift the inverse Laplace contour to $\operatorname{Re} z = -\delta$ and integrate the $|b|^{-2}$ bound to obtain exponential L^1 decay. Fixed-domain resolvent differentiation and the analytic source theorem justify parameter and source derivatives under the contour integral. $\square$

Theorem 1.10 (Round-twenty-one A4 closure). *The exact Sinai history platform has derived exponential variation, Lyapunov/minorization estimates, a proved weighted-Lipschitz spectral gap, a fixed-space analytic Feynman–Kac pressure, an enhanced suspension limit, an operator renewal theorem, and a finite resolved-space memory kernel with exponential decay.*

Proof. Combine Theorems 1.4, 1.5, 1.6, 1.7, and 1.9. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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